

A Multi-Stage Machine Learning Approach for Stock Price Prediction: Engineered and Derivative Indices

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Abstract

In this paper, a multi-stage machine learning approach is proposed to predict the next day's stock prices. The proposed approach has three main stages, i.e., data collection, feature space generation, and prediction. In the first stage, 5283 records of daily historical series of open prices, close prices, the highest prices, the lowest prices, and the trading volume of assets of four well-known stocks are selected from FTSE 100. In the second stage, a novel set of engineered and derivative indices extracted from the original time series for stock price prediction are proposed. In the third stage, two Multi-Layer Perceptron (MLP) are proposed to predict the next day's stock prices using the engineered discrete and continuous indices. The case study uses the daily historical time series of stock prices between January 1, 2000, and December 31, 2020. The proposed machine learning approach presents suitable applicability and accuracy, respectively.

Keywords:

Machine Learning; Artificial Neural Network; Stock prediction; Price Time series; Financial Forecasting.

1. Introduction

Stock price prediction is one of the most challenging problems in the field of financial time series due to its dynamism, nonlinearity, and chaotic nature, in which many parameters often change randomly (Ma et al., 2021; Chandar et al., 2016). Stock price prediction can be considered as an essential task for investors and financial institutions, because accurate prediction of the stock price can lead to suitable trade-off among return and risk of investment (Chen et al., 2021; Tan et al., 2019).

Portfolio optimization has turned into an important and challenging issue in the field of finance and investment. Portfolio optimization is a good combination of stocks such that the profit of one stock can offset the loss of other stocks (Mansini et al., 2015). Investors are usually interested in knowing the future status of the stocks they have invested in, and the measures that help them to form the best portfolio (Wang et al., 2020). Accordingly, stock selection can be considered as a prerequisite and preliminary step for the portfolio optimization. Therefore, combination of portfolio optimization and forecasting theory in the form of using accurate models to predict the stock prices and trends can help the investors to pre-select the proper stocks with potential for greater profits in order to form an optimal portfolio (Yang et al. 2019; Ma et al., 2021). Due to the importance of the stock market prediction, many researchers have addressed the issue including stock prices, stock returns, market indices and stock price trend. Although there are several researches in the related field, due to our best knowledge there is no certain study in which a complete set of derivative and engineered indices are used to predict the stock price time series rather than direct forecasting of data.

The main purpose of this paper is to develop a multi-stage machine learning approach to predict the next days' stock prices using artificial neural networks (ANNs) through a novel and complete set of engineered and derivative indices. On the other hand, in this paper we will extract derivative indices from original stock price time series. We will predict the stock prices

using these set of derivative indices rather than applying original stock price time series. The proposed approach of this paper has several innovative aspects as follows:

- A wide variety of input variables has been considered in this study.
- Engineered and derivative indices for stock price prediction have extracted from daily stock price time series data. These indices are to be considered as target variables. These targets are:
 - o (1) the next day trend;
 - o (2) whether the next day will be a peak?
 - o (3) whether the next day will be a trough?
 - o (4) prediction of the distance between two consecutive peaks
 - o (5) prediction of the distance between two consecutive troughs
 - o (6) prediction of the distance from one peak or trough to the next.
 - o (7) prediction of the height of the next peak
 - o (8) prediction of the depth of the next trough
 - o (9) comparing the closing price of two consecutive trading day
 - o (10) the next day stock return
- To our best knowledge, the composition of these variables has not been used in previous researches. We are trying to extract a deep knowledge from the time series of stock price using these engineered indices. Combination of these indices will probably cover different hidden and complicated aspects of stock price time series based on a new perspective.
- It has been applied on a real case study including 5283 records of four stock prices from FTSE 100. So, the proposed model is a practical approach which can be used by many investors in order to automate short-term investment decision-making.

The paper is organized as follows. In Section 2, a literature review of past research works is presented. In Section 3, research methodology is presented. In Section 4, empirical results are represented. In Section 5, conclusions are presented.

2. Literature of Past Works

Successful prediction of price in the stock market has a significant impact on investors' profits. So, it has attracted the attention of many researchers. This section provides an overview of previous researches in the related fields.

Ma et al., (2021) applied random forest (RF), support vector regression (SVR), LSTM neural network, deep multilayer perceptron (DMLP) and convolutional neural network for the next day return prediction. Ribeiro et al. (2021) proposed a novel hybrid model named HAR-PSO-ESN that combines heterogeneous autoregressive (HAR) specification, the echo state network (ESN), and the particle swarm optimization (PSO) metaheuristic for predicting realized volatility of stock prices. Kumar Chandar (2021) proposed nine new integrated models for forecasting intraday stock price based on the potential of three ANNs.

Paiva et al., (2019) proposed a hybrid model using support vector machine (SVM) and mean-variance (MV) method for portfolio selection. SVM algorithm applied for predicting whether a stock has the potential to gain expected return (1% return in $t+1$) or not. Zhang, et al., (2020) proposed a hybrid deep learning model to predict the one-step-ahead closing price of stock indices. Vijh et al., (2020) applied machine learning techniques such as artificial neural networks (ANN) and Random Forest (RF) for predicting the next day closing price of the stock. Yang et al., (2019) proposed a novel hybrid stock selection method. The authors applied extreme learning machine (ELM) in order to implement stock price prediction. Bisoi et al., (2019) addressed stock price and trend prediction problems and using the combination of robust kernel extreme learning machine (RKELM) and variational mode decomposition

(VMD) for this purpose. [Ismail et al., \(2020\)](#) addressed the next day direction of stock price movement problem. The classification was implemented using logistic regression (LR), artificial neural network (ANN), support vector machine (SVM) and random forest (RF) algorithms. [Tan et al., \(2019\)](#) implement the RF algorithm in the field of stock selection strategy. Several fundamental/technical variables as well as momentum have been used to predict long-term and short-term stock price trend. [Zhou et al., \(2019\)](#) integrated Empirical Mode Decomposition (EMD) and Factorization Machine (FM) based Neural Network technique for stock market trend prediction. [Basak et al., \(2019\)](#) applied tree-based classifiers i.e. RF, and gradient boosted decision trees for predicting the direction of stock market prices.

[Xu et al., \(2020\)](#) proposed time series predictive model using integration of wavelet transform, the stacked auto-encoders and the bidirectional long-short term memory. Discrete wavelet transform was applied to de-noise the stock data, stacked auto-encoders were applied to extract deep features from technical indicators. The bidirectional long-short term memory (LSTM) applied as forecasting algorithm. A summary of previous studies in the field of stock prediction are listed in [Table 1](#).

Table 1: Selected Stock Prediction literature

Researcher	Dataset	Feature set		Applied Method
		Input Variables	Target	
Ma et al., (2021)	49 stocks from China Securities 100 Index	Past 60 days' daily returns	Next day's return	DMLP, LSTM neural network and CNN, random forest, support vector regression
Ribeiro et al. (2021)	Three Nasdaq companies: Caterpillar (CAT), eBay (EBAY), and Microsoft (MSFT).	Past values of series	Realized volatility of stock prices	Echo state network Heterogeneous autoregressive model Particle swarm optimization
Kumar Chandar (2021)	SBIN.NS, NTPC.NS, TATASTEEL.NS, INFY.NS	Opening price, highest price, lowest price and closing price, several technical indicators	Intraday stock price	Nine hybrid model based on ANN
Zhang et al., (2020)	SHCOMP, SZSE, GEM, HIS, DJIA, S&P 500 Index	Extracted Intrinsic Mode Functions (IMFs)	One-step-ahead closing price	LSTM
Ismail et al., (2020)	Kuala Lumpur Composite Index (KLCI)	Input vectors of invariant topological features	Next day direction of stock price movement	LR, ANN, SVM, and RF
Vijh et al., (2020)	Nike, Goldman Sachs, Johnson and Johnson, Pfizer and JP Morgan Chase and Co.	H-L, O-C, 7 DAYS MA, 14 DAYS MA, 21 DAYS MA, 7 DAYS STD DEV	next day closing price	ANN, Random Forest
Tan et al., (2019)	Chinese stock market	Fundamental/Technical variables as well as momentum	long-term and short-term stock price trend	Random forest (RF)
Yang et al., (2019)	China's A-share stock market	Close, open, high, low, average price, MCAP, return rate, volume, turnover, turnover rate, volatility, general capital, P/E, P/B, PC/F, and P/S.	Price, Return	Extreme learning machine

Bisoi et al., (2019)	BSE S&P 500 Index, Hang Seng Index, and Financial Times Stock Exchange 100 Index	Lagged intrinsic mode functions (IMFs) and lagged modes	Price, Trend	Robust kernel extreme learning machine
Paiva et al., (2019)	São Paulo Stock Exchange Index	Return lagged based on opening, closing, maximum, and minimum prices as well as several technical variables such as Momentum, RSI, ATR, Chaikin A/D line, on balance volume	Determine the stock has the potential to gain 1% return in t+1	SVM
Zhou et al., (2019)	NASDAQ, S&P 500	Extracted Intrinsic mode functions (IMFs) from Stock Close Price.	Determining the changes (directions) and predicting the future prices (trends) in stock prices.	Neural Networks
Basak et al., (2019)	AAPL, AMS, AMZN, FB, and MSFT	RSI, SO, %R, MACD, PROC, OBV	Direction of stock market prices	random forests, and gradient boosted decision trees
Lee and Yoo (2018)	10 stocks in the S&P500	open, high, low, close, and volume	Return	RNN, GRU, LSTM
Lei (2018)	SSE Composite Index, CSI 300 Index, All Ordinaries Index, Nikkei 225 Index, and Dow Jones Index	Random % K, Random % D, W&R, MTM, RSI, BBI, 5BIAS, 10BIAS, MB, PSY, AR, BR, MV, TAPI, VR	stock price trend	Wavelet Neural Network, Rough Set
Hiransha et al., (2018)	National Stock Exchange of India, and New York Stock Exchange	Daily closing price	stock price	MLP, RNN, LSTM, and CNN
Chen and Hao (2018)	Shanghai and Shenzhen stock exchange markets	Open, high, low, and close prices. Technical indicators such as: MA, EMA, MACD, RSI, OBV, MoM, ATP ¹³ , ALT ¹⁴ , ITL ¹⁵	Stock trading signal classification (considering four level for target including strongly recommended buying signal (SBS), ordinarily recommended buying signal (OBS), ordinarily recommended selling signal (OSS), and strongly recommended selling signal (SSS).	Weighted SVM
Abe and Nakayama (2018)	Japanese stock market	Fundamental Features	One-month-ahead stock returns	
Fu et al., (2018)	Chinese stock data	Technical and Fundamental Features	Labeling each stocks according to their ranking with respect to the return-to-volatility ratio	Logistic Regression, RF, DNN
Weng et al., (2017)	AAPL, NASDAQ	Open, high, low, and close prices, and volume of trades, RSI ¹⁰ , P/E ¹⁶ , %K ¹⁷ , %R ¹⁸	Stock Market One-Day Ahead Movement: 1: Open (i + 1) – Close(i) 2: Open (i + 1) – Open(i) 3: Close (i + 1) – Close(i) 4: Close (i + 1) – Open(i) 5: Trade Volume (i + 1) – Trade Volume(i)	ANN, DT, SVM
This Study	Randomly chooses four stocks from FTSE 100	Several lagged variables of returns, several lagged variables of trading volume changes, several technical indicators, Extracted sub-series from decomposition	Trend, Peak Detection (PK), Trough Detection (TRU), Two Consecutive Peaks (DBP), Distance Between Two Consecutive Troughs (DBT), Distance from One Peak or Trough to The Next (DBPT), Peak Height (PKHe), Trough Depth (TRUDe), Distance Between, Delta, Rate of Return (Ret.)	Multi-Layer Perceptron (MLP)

Statistical methods such as auto-regression moving average (ARMA) (Baek and Kim, 2018), autoregressive integrated moving average (ARIMA) (Wang et al., 2020; Chen et al., 2021), Autoregressive conditional heteroscedasticity (ARCH) (Chen et al., 2021; Gandhmal, and Kumar, 2019), generalized autoregressive conditional heteroskedasticity (GARCH) (Chen et al., 2021; Gandhmal, and Kumar, 2019), and the smooth transition autoregressive model (STAR) (Zhou et al., 2019) have been previously applied to stock market prediction.

There are several studies that show the superiority of machine learning (ML) methods over classical methods in terms of financial time series predictions (Sezer et al., 2020; Freitas, et al., 2009). Many researchers claimed that machine learning methods deal with the complex, nonlinear, multi-dimensional and noisy financial time series better than traditional statistical models (Wang et al., 2020; Baek and Kim, 2018; Chen et al., 2021). Different kind of ML methods such as support vector machines (SVM) (Lee, 2009; Paiva et al., 2019; Chen and Hao, 2018), support vector regression (SVR) (Lahmiri, 2018; Ma et al., 2021) neural networks (NNs) (Chen et al., 2021; Zhou et al., 2019; Ismail et al., 2020), Naive-Bayes (Patel et al., 2015), K-Nearest Neighbor (Ballings et al., 2015; Subha, Nambi, 2012), ensemble learning i.e. Random forest (RF) (Wang et al., 2020; Tan et al., 2019; Ballings et al., 2015; Patel et al., 2015), AdaBoost (AB) (Ampomah et al., 2020; Ballings et al., 2015), Bagging (Ampomah, 2020) have been applied on stock market prediction.

In the last few years Deep Learning (DL) as a subfield of ML has been introduced by many researchers as one of the best and most powerful methods of financial time series forecasting. Different type of deep learning algorithms including Deep Multi-Layer Perceptron (DMLP) (Sezer et al., 2017), Recurrent Neural Network (RNN) (Pawar et al., 2019), Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNNs) have been widely applied on this subject (Sezer et al., 2020).

In order to effectively predict the future status of financial time series, in addition to choosing a strong forecasting method, selecting appropriate indicators and input attributes will play a significant role (Wang et al., 2020; Long, Lu, Cui, 2019). Most studies only used raw price data (Open, High, Low, and Close prices, and trading volume of assets) as the input to prediction models (Lee and Yoo, 2020; Chen et al., 2015; Dezsi and Nistor, 2016). Many studied applied several lagged variables of return (Wang et al., 2020; Paiva et al., 2019; Ma et al., 2021). Some studies have used a variety of technical indicators as input (Dai et al., 2021; Basak et al., 2019). Others have used fundamental variables (Abe and Nakayama, 2018). Some studies have used a combination of technical and fundamental indicators (Tan et al., 2019; Fu et al., 2018).

3. Proposed Multi-Stage Machine Learning Approach for Stock Price Prediction

The structure of proposed machine learning approach is introduced in this section.

3.1. Proposed Multi-Stage Machine Learning Structure

As mentioned, the main purpose of this paper is to develop a multi-stage machine learning approach to predict the next days' stock prices using artificial neural networks (ANNs) through a novel and complete set of engineered and derivative indices. The proposed conceptual model, as shown in Figure 1, is composed of three main stage.

In the first stage, data collection and in the second stage feature space generation are done. In the third stage, the prediction using Multi-Layer Neural network are handled. In the following, the techniques and algorithms used in the proposed multi-stage machine learning approach for stock price prediction are briefly described.

3.2. Conceptual Structure of proposed multi-stage machine learning approach

Figure 1 presents the main stages, i.e., data collection, feature space generation, and prediction for the multi-stage machine learning approach for stock price prediction.

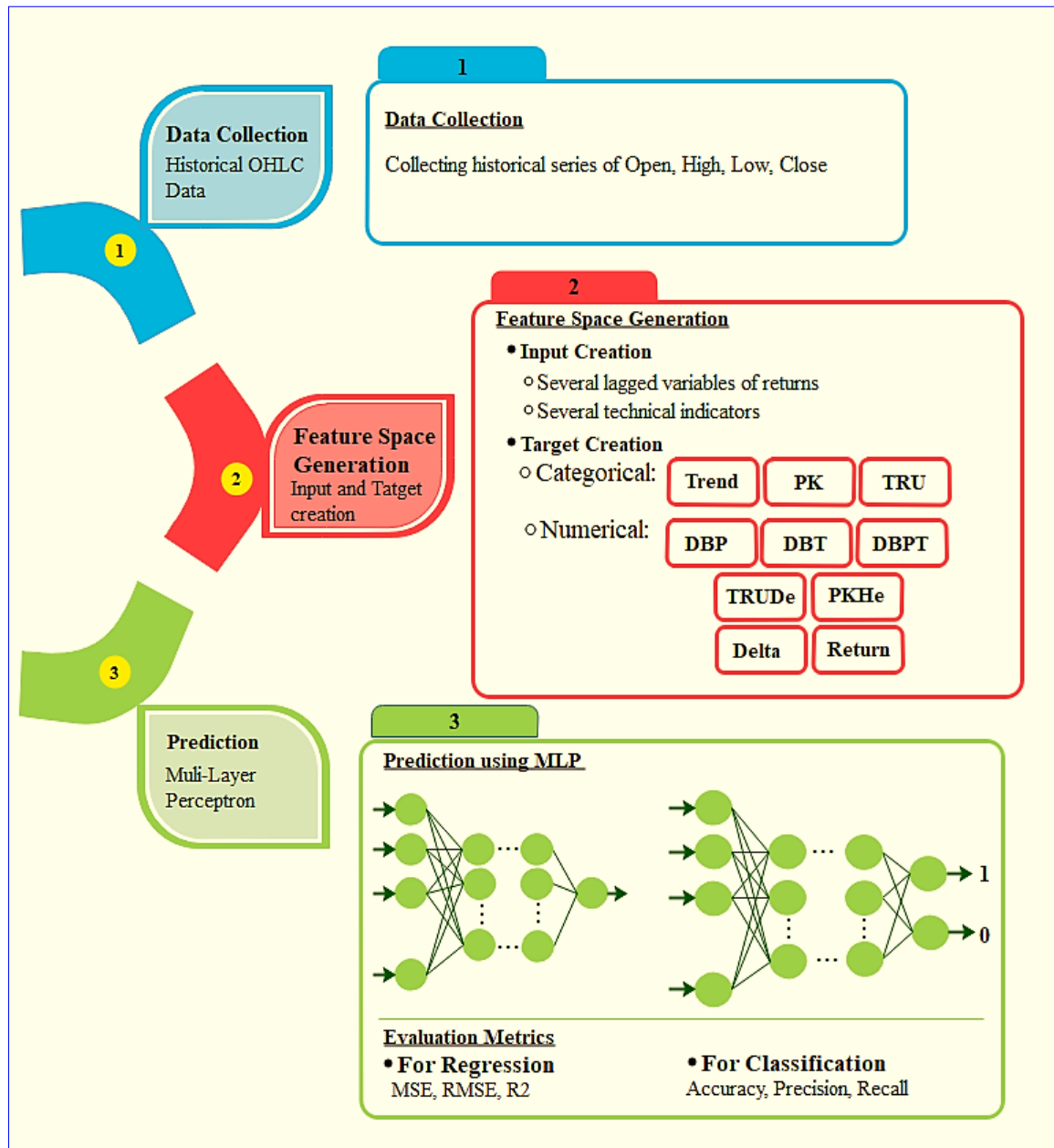


Figure 1: Proposed Methodology of the Research

3.2.1. First stage: Data Collection

Four well-known stocks (including: “BP p.l.c.” (BP), “BHP Group Ltd.” (BHP), “GlaxoSmithKline” (GSK), and “Vodafone Group Plc” (VOD) were selected from FTSE 100.

Daily stock price of these stocks for the period from January 1, 2000 until December 31, 2020 were gathered from yahoo finance and considered as datasets of this research. The abbreviations of the selected stocks are used for sake of brevity in the following sections. The initial data involves the daily historical series of open prices, close prices, the highest prices, the lowest prices, and the trading volume of assets. A vector of five dimensions with 5283 records of daily stock prices were considered for each stock.

3.2.2. Second stage: Feature Space Generation

As mentioned, in order to make a prediction of stock price, a complete set of engineered and derivative indices are extracted from original stock price time series in this research. In this section the proposed engineered indices, their descriptions and the method of mapping original data set into the considered space are presented.

• Input Creation

The selection of input variables is one of the important issues in implementing stock price prediction approaches. We considered both lagged variables of return and technical indicators as inputs. The input variables used in this study are presented in [Table 2 \(Paiva et al., 2019; Wang et al., 2020\)](#).

Table 2: Input variables used in this study

Row	Attribute	Row	Attribute
1	—	15	-
2	-	16	Relative Strength Index (RSI)
3	-	17	Momentum
4	-	18	True Range (TR)
5	—	19	Average True Range (ATR)
6	—	20	parabolic SAR
7	—	21	Commodity Channel Index (CCI)
8	—	22	Simple Moving Average (SMA)
9	-	23	Weighted Moving Average (WMA)
10	-	24	Hull Moving Average (HMA)
11	-	25	Average directional index (ADX)
12	—	26	William's % R.
13	-	27	% K
14	-	28	% D-slow

The technical indicators used in this research and their calculation formula are presented in [Table 3](#).

Table 3: Selected Technical indicators and their formula

Technical Indicator	Attribute	References
Simple n-day moving average (here n=14)	—	Kara et al., (2011)
WMA (Weighted n-day moving average)	—	Kara et al., (2011)
HMA (Hull Moving Average)	/	Gorgulho et al., (2011)
Momentum		Kara et al., (2011)
Stochastic %K	—	Kara et al., (2011)
Stochastic %D	—	Kara et al., (2011)
Slow %D	—	Kara et al., (2011)
Larry William's R%	—	Kara et al., (2011)
RSI (Relative Strength Index)	—	Kara et al., (2011)
CCI (Commodity Channel Index)	—	Kara et al., (2011)
TR (True Range)		Bruni (2017)
ATR (Average True Range)	—	Bruni (2017)
Average Directional Index (ADX)	—	Bruni (2017)
Parabolic SAR		Wilder (1978)

is the closing price at time , is the low price at time , is the High price at time , and mean lowest low and

highest high in the last , days , means the upward price change, means the downward price change at time . —

,

— ,

denotes the exponential moving average of X over the last n periods

value represents the acceleration factor. EP is the extreme point in a trend (highest point reached by a price The during an uptrend or the lowest price reached during a downtrend).

• **Target Creation**

In this section, a set of novel engineered indices is derived from daily stock price data to be tested as the target. Attributes entitled “Trend”, “Peak Detection (PK)”, “Trough Detection (TRU)”, “Distance between two consecutive peaks (DBP)”, “Distance between two consecutive Troughs (DBT)”, “Distance from one peak or trough to the next (DBPT)”, “Height of the next peak (PKHe)”, and “Depth of the next trough (TRUDe)”, “Delta”, and “Daily Return (Ret.)” are developed based on original stock close price time series. The

calculated attributes are then used separately as target variable in order to explore the future pattern of price time series. These features are presented in Table 4.

Table 4: Set of Derivative indices that be tested as Target variables

Row	Attribute	Description	Type
1	Trend	Stock price movement	Binomial (0,1)
2	PK	Peak detection	Binomial (0,1)
3	TRU	Trough detection	Binomial (0,1)
4	DBP	Distance between two consecutive peaks	Numeric
5	DBT	Distance between two consecutive troughs	Numeric
6	DBPT	Distance from one peak or trough to the next	Numeric
7	PKHe	Height of the next peak	Numeric
8	TRUDe	Depth of the next trough	Numeric
9	Delta	The difference between the closing price of the current period and the previous period	Numeric
10	Ret.	Daily return of the selected stocks	Numeric

The method of calculating each engineered index is described as follows. In all equations refers to the number of records of sample dataset per each stock.

- **Trend:** This index indicates the direction of an asset's price. In this research, “Trend” is considered as a binomial variable. If P_t denotes the closing price of stock in period t , so that for each day if the closing price of the current period is greater than the closing price of the previous period, “Trend” it is equal to one, otherwise it is equal to zero.

(1)

- **Peak Detection:** Peak Detection is a binominal variable that indicates whether each point in the dataset is a peak or not. The peak is the point at which the uptrend has turned into downtrend. This index is abbreviated to PK for convenience in the following. PK is defined by Eq. (2).

(2)

- **Trough Detection:** Trough Detection is a binominal variable that indicates whether each point in the dataset is a Trough or not. The Trough is the point at which the downtrend has turned into uptrend. This index is abbreviated to TRU for convenience in the following. “TRU” is defined as Eq. (3).

(3)

- **Distance Between Two Consecutive Peak:** This index indicates the distance between two consecutive peaks. If P denotes the number of peaks in the sample dataset and i represents the i peak, then the distance between the i and j peaks is calculated using Eq. (4). This index is abbreviated to “DBP” for convenience in the following

(4)

where t_i is denote the time period of occurrence of each peak.

- **Distance Between Two Consecutive Trough:** This index indicates the distance between two consecutive troughs. If T denotes the number of troughs in the sample dataset and i represents the i trough, then the distance between the i and j troughs is calculated using Eq. (5). This index is abbreviated to “DBT” for convenience in the following.

(5)

where t_i is denote the time period of occurrence of each peak.

- **Distance from one peak or trough to the next:** This index indicates the distance from one peak to its next trough, or the distance from one trough to its next peak. If n and m denotes the number of peaks and troughs respectively, p_n represents the n peak and t_m represents the m trough, then the distance between p_n and t_m is calculated using Eq. (6). This index is abbreviated to “DBPT” for convenience in the following.

(6)

where

- **Height of the Next Peak:** This index indicates the heights of a peak. If p_n denotes the minimum of the close price of sample dataset and p_{n+1} denotes the close price of the peak p_n , and t_m in this case, the height of p_n is calculated using Eq. (7). This index is abbreviated to PKHe for convenience in the following.

(7)

- **Depth of the Next Trough:** This index indicates the Depth of a trough. If p_n denotes the maximum of the close price of sample dataset and p_{n+1} denotes the close price of the peak, t_m , in this case, the height of t_m is calculated using Eq. (8). This index is abbreviated to TRUDe for convenience in the following.

(8)

- **Delta:** Delta represents the difference between the closing price of two sequential periods. Delta in this study is defined by Eq. (9).

(9)

where p_n denotes the closing price of stock in period n .

- **Daily Return:** This index represents the stock daily return and measures the changes in a stock’s price as a percentage of the previous day’s closing price. This index is abbreviated to “Ret” for convenience in the following.

—

(10)

Figure 2 also represents the schematic view and calculation method of proposed indices.

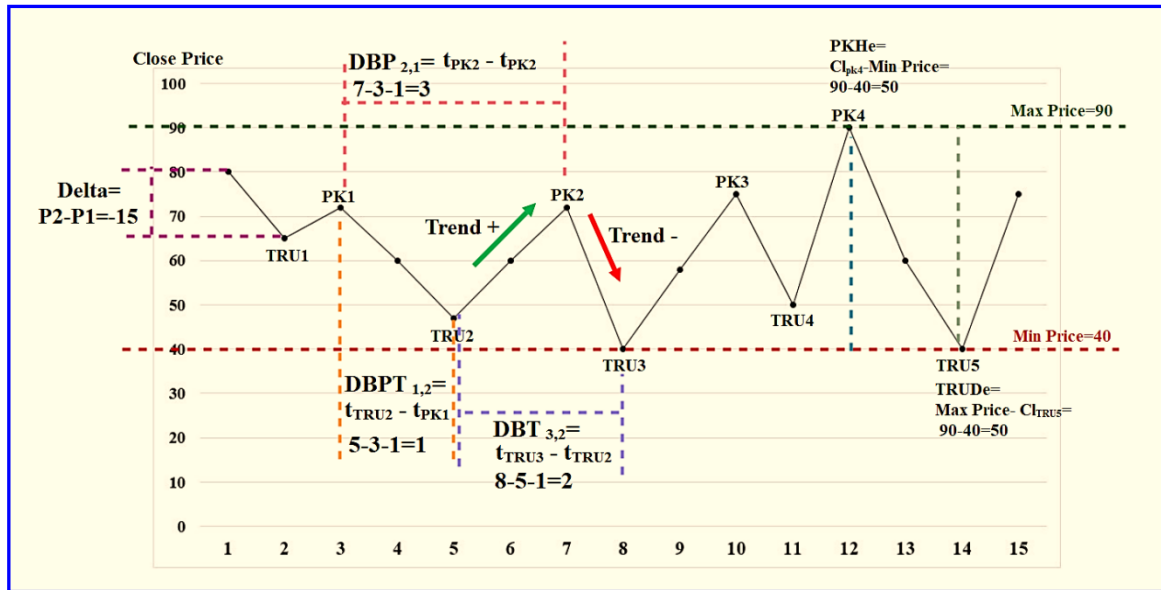


Figure 2: Calculation method of the proposed indices from closing price

3.2.3. Third stage: Prediction

The derivative indices for all selected stocks for the next period are predicted thereby serving to reflect the future features of price time series and price prediction.

For ten derived indices including “Trend”, “PK”, “TRU”, “DBP”, “DBT”, “DBPT”, “PKHe”, “TRUDe”, “Delta”, and “Ret.” the prediction function for the next period is presented as follows.

$$\hat{Y}_{t+1} = f(X_t) \quad (11)$$

Therefore, these indices are estimated based on its associated inputs data X_t . The Y_{t+1} represents the $t+1$ element of the sample dataset for period t . The set of predictor variables was represented in Table 3.

3.2.3.1. Prediction using Multi-Layer Perceptron

Artificial Neural Networks (ANN) is one of the most used methods in time series forecasting (Erzurum Cicek, and Kamisli Ozturk, 2021). Multi-layer Perceptron (MLP) is a kind of artificial neural network which is relatively easy to implement. MLP has the ability to learn the complex relationship between the input variables and target due to the presence of activation function in each layer.

The following general objectives may be considered in multilayer perceptron training:

- **Regression.** In this case, the network regresses the independent variables, provided by the inputs, onto the dependent variable. The input variables can be categorical or numeric types, however, for regression ANNs, the dependent variable is numerical. The regression method will produce an estimated value of a dependent variable, which minimizes the mean-squared error.

- **Supervised classification.** In this case, the output variable is a categorical variable (or binary) and an input vector or pattern is to be trained to select a class from a list of classes.

The general topology of MLP consists of an input layer, one or more hidden layers, and an output layer. In the input layer, a neuron is considered for each of the input variables. The hidden layer is used to show the nonlinear relationship between the variables and completes the data processing. Each hidden layer can have any number of neurons. The output layer is

producing the final result. The output layer is aimed to produce the categorical class label or is used to predict continuous target indices. In Regression cases, the output layer will have only one neuron. In classification cases, the output layer should have neurons exactly the same number of classes of the target variable. (Siswanto et al., 2016; Huang et al., 2004; Murtagh, 1991)

The schematic architecture of MLP for regression and classification problems is presented in Figure 3.

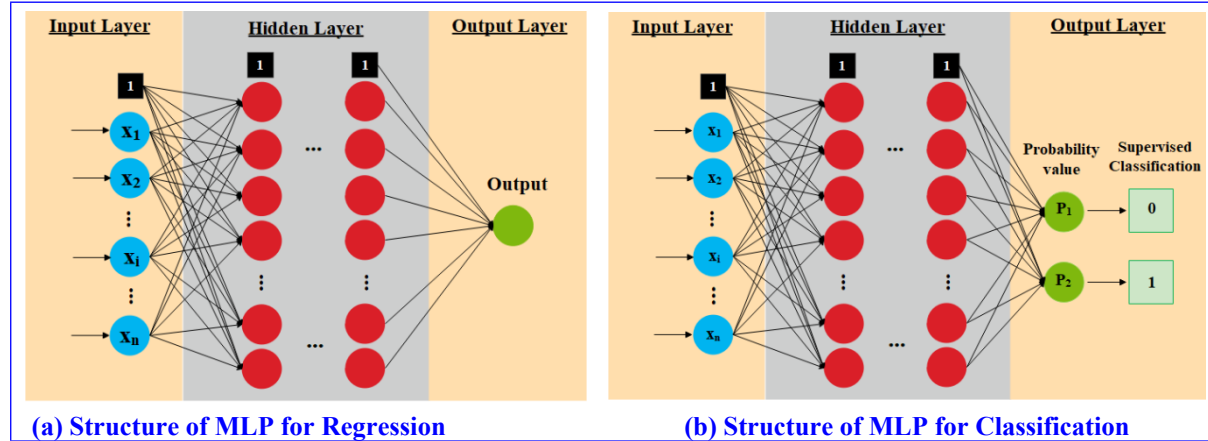


Figure 3. ANNs' Structures

The structure of a neuron consists of inputs , weights , bias b , transfer function , and output is presented in Figure 4.

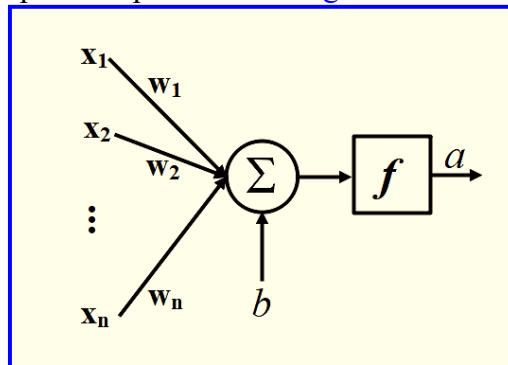


Figure 4: Schematic view of a neuron

The neuron inputs come from the other neurons in the previous layer. All weighted inputs and biases are summed and the result is passed through the transfer function to generate output, as in Eqs. (1) and (2). The output is sent to other neurons in the next layer as input.

(12)

(13)

A hidden layer receives input from the previous layer and provides output to another hidden layer or an output layer. Various nonlinear activation functions such as Rectified Linear Activation (ReLU), Logistic (Sigmoid), Hyperbolic Tangent (Tanh) can be used in the hidden layers of a neural network. The activation function in the output layer must be selected according to the type of target variable. In a regression problem, a linear activation function should be used. In classification problems Soft-max, and sigmoid activation

functions can be used. If the activation function is Sigmoid or Soft-max, the probability value is received from all nodes of the output layer. For the training process of ANNs, several learning techniques such as Levenberg-Marquardt (LM) and Scaled Conjugate Gradient (SCG) can be used for regression and classification problems respectively. The Mean Squared Error, or MSE, is the loss that can be used for regression problems and Cross-entropy is the loss function that can be used for classification problems.

- **Evaluation Metrics**

Evaluation metrics for prediction models: To evaluate the performance of continuous prediction models, three metrics including mean squared error (MSE), root mean squared error (RMSE), and R squared (R^2) are applied. These metrics are defined as follows.

Mean square error (MSE): It is an indicator measuring the average squared difference between the observed values and predicted values (Ma et al., 2021; Vijn et al., 2020).

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (14)$$

Root-mean-square error (RMSE): It is another effective indicator measuring differences between the observed values and predicted values.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (15)$$

Coefficient of determination (R^2): This is a measure of how well the model can be explained.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (16)$$

Evaluation metrics for classification models: In order to evaluate classification models three metrics i.e. as Accuracy, Precision, and Recall are used. These metrics are calculated using True Positive (TP), False Positive (FP), True Negative (TN) and False Negative (FN). The formula of these metric is depicted in Equations (17)– (19) (Patel et al., 2015).

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (17)$$

$$Precision = \frac{TP}{TP + FP} \quad (18)$$

$$Recall = \frac{TP}{TP + FN} \quad (19)$$

4. Empirical Results

The results of proposed MLP for classification and prediction of designed engineered and derivative indices have been presented in this section.

4.1. Structure of Proposed MLP

The proposed MLP is implemented in Matlab R2018b in this study. Two distinct Multi-layer perceptron have been used to perform prediction of continuous indices and classification of binary variables. “Trend”, “PK”, and “TRU” are binary target indices and “Delta”, “Return”, “PKHe”, “TRUDe”, “DBP”, “DBT”, and “DPBT” are continuous target indices. Each MLP was employed for each target and all four selected stocks. All variables which were presented in Table 2 are assumed as input of both MLPs. Data were divided for training, testing, and validation by 70%–15%–15%, respectively. This division is the most well-known setting for portioning the data patterns. The hidden layer was tested using 5, 10, 15, 20, 25, 30, and 35 neurons for both MLPs, and the best setting was selected.

For categorical indices Tangent sigmoid (Tansig) was used as an activation function at hidden layer nodes and 'Soft-max' was used as a linear transfer function at the output layer.

Tangent sigmoid (Tansig) was used as an activation function at hidden layer nodes and ‘Purelin’ was used as a linear transfer function at the output layer for continuous target indices.

Levenberg–Marquardt (LM) method was used as training algorithm for prediction models and Scaled Conjugate Gradient (SCG) was used as training algorithm for prediction models. The parameters of the both MLPs are briefly shown in Table 5.

Table 5: The MLP model parameters setting

Parameter	Classification	Prediction
Number of the input layer neurons	28	28
Number of neurons in hidden layer	5, 10, 15, 20, 25, 30, 35	5, 10, 15, 20, 25, 30, 35
Number of epochs	500, 1000	500, 1000
Performance Goal	0	1e-8
Minimum Gradient	1e-06	1e-07
Maximum Validation Checks	6	20
Transfer function	tansig, softmax	tansig, purelin
Training function	Scaled Conjugate Gradient (SCG)	Levenberg–Marquardt training algorithm
Performance function	Cross-Entropy	MSE

The results of ten target derivative indices are presented in the following. Modeling was performed for all 4 selected stocks separately.

First, the results of classification of binomial indices including ‘Trend’, ‘PK’, and ‘TRU’ are presented.

4.2. Trend Classification

The results of closing price trend classification are presented in Table 6. As trend is a binary target, the performance of the classification MLP is evaluated using Accuracy, Precision, and Recall. As can be seen, the trend forecasting has been done with high accuracy.

Table 6: The results of trend classification using ANN model

Stock	Dataset	Evaluation Metrics		
		Accuracy	Precision	Recall
BP	All Data	97.7	98.0	97.3
	Train	98.0	98.0	98.0
	Test	97.0	98.2	95.8
	Validation	97.0	98.2	95.7
BHP	All Data	96.9	96.7	96.9
	Train	97.0	96.9	97.0
	Test	95.8	95.2	95.9
	Validation	97.1	96.9	97.2

GSK	All Data	96.7	96.7	96.8
	Train	96.9	96.9	97.0
	Test	96.8	97.8	96.1
	Validation	95.8	95.0	96.2
VOD	All Data	96.1	96.4	96.0
	Train	96.4	96.6	96.4
	Test	96.6	96.7	96.5
	Validation	94.3	95.5	93.9

4.3. Peak Detection (PK)

As mentioned, the “PK” variable indicates whether the closing price will be at the peak point the next day or not. In fact, PK is a binary variable, where PK=1 if the next day is peak and PK= 0 otherwise. The results are shown in Table 7. As can be seen the value accuracy, precision and recall metrics for all selected stocks are suitable.

Table 7: The results of peak detection using ANN model

Stock	Dataset	Evaluation Metrics		
		Accuracy	Precision	Recall
BP	All Data	75.5	88.6	80.6
	Train	75.3	88.1	80.7
	Test	76.6	90.3	81.1
	Validation	75.3	89.3	80.0
BHP	All Data	75.6	93.0	78.4
	Train	75.4	93.0	78.1
	Test	75.9	93.9	78.2
	Validation	76.6	92.2	80.1
GSK	All Data	76.2	88.8	81.1
	Train	75.1	88.2	80.0
	Test	80.6	91.2	84.9
	Validation	76.6	88.8	82.2
VOD	All Data	75.3	88.8	80.0
	Train	75.9	88.5	81.1
	Test	73.6	89.3	77.2
	Validation	74.2	89.7	78.2

4.4. Trough Detection

As mentioned, the “TRU” variable indicates whether the closing price will be at the trough point the next day or not. In fact, TRU is a binary variable, where if TRU=1 the next day is trough and if TRU = 0 otherwise. The performance of the proposed ANN is represented in Table 8.

Table 8: The results of trough detection using ANN model

Stock	Dataset	Evaluation Metrics		
		Accuracy	Precision	Recall
BP	All Data	75.0	93.4	77.7
	Train	74.6	93.1	77.4
	Test	75.6	95.1	77.5
	Validation	76.4	93.3	79.7
BHP	All Data	76.1	87.3	81.9
	Train	76.7	88.0	82.1
	Test	75.1	85.8	82.2

GSK	Validation	74.4	85.7	80.7
	All Data	76.0	89.1	80.8
	Train	76.2	88.9	81.0
	Test	75.6	89.1	80.4
VOD	Validation	75.6	89.9	80.2
	All Data	74.3	87.0	80.0
	Train	74.6	87.2	80.3
	Test	74.2	87.3	79.5
	Validation	72.6	85.5	79.0

In the following, the results of prediction of numerical indices including ‘DBP’, ‘DBT’, ‘DBPT’ ‘PKHe’, ‘TRUDe’ ‘Delta, and ‘Ret.’, are presented.

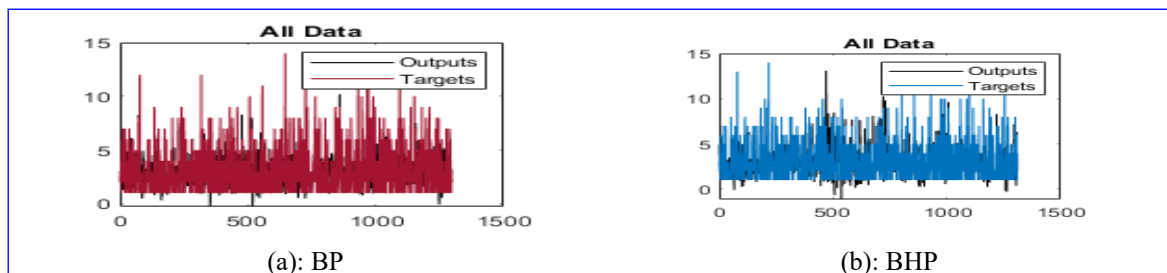
4.5. DBP Prediction Results

The results of predicting the distance between two consecutive peaks (DBP) is presented in Table 9. The R^2 values are greater than 0.22 for all stocks in train, test, and validation data.

Table 9: The results of DBP prediction using ANN model

Stock	Dataset	Evaluation Metrics		
		MSE	RMSE	R^2
BP	All	2.7741	1.6656	0.56291
	Train	2.1137	1.4539	0.65554
	Test	3.6385	1.9075	0.41247
	Validation	4.9816	2.232	0.35844
BHP	All	2.6967	1.6422	0.58192
	Train	2.3249	1.5248	0.656
	Test	3.2445	1.8013	0.4837
	Validation	3.8777	1.9692	0.34357
GSK	All	3.0307	1.7409	0.52903
	Train	2.3638	1.5375	0.65273
	Test	4.9127	2.2165	0.35139
	Validation	4.2529	2.0623	0.21827
VOD	All	2.3874	1.5451	0.55254
	Train	1.836	1.355	0.67592
	Test	2.8198	1.6792	0.4462
	Validation	4.5366	2.1299	0.22756

Figure 5 presents the real target and the output of ANN for “DBP” in four selected stocks. The differences between target and output can be detected.



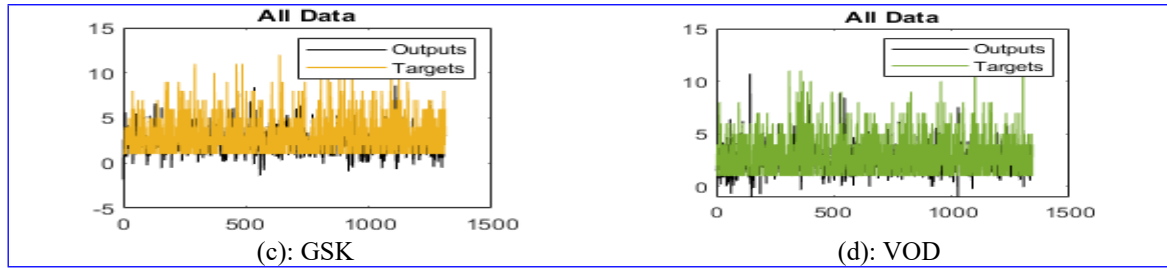


Figure 5: Comparing the actual and the predictions of the ANN model for DBP

4.6. DBT Prediction Results

The results of predicting the distance between two consecutive troughs (DBT) are presented in Table 10.

Table 10: The results of DBT Prediction using ANN model

Stock	Dataset	Evaluation Metrics		
		MSE	RMSE	R ²
BP	All	2.9245	1.7101	0.51853
	Train	2.3042	1.5179	0.62628
	Test	3.6571	1.9124	0.3878
	Validation	5.077	2.2532	0.31447
BHP	All	2.7854	1.669	0.53692
	Train	2.7102	1.6463	0.57008
	Test	3.1475	1.7741	0.53562
	Validation	2.773	1.6652	0.42146
GSK	All	2.9393	1.7144	0.45242
	Train	2.6199	1.6186	0.4939
	Test	4.5269	2.1277	0.29305
	Validation	2.8401	1.6853	0.46125
VOD	All	2.2899	1.5132	0.54817
	Train	2.1655	1.4716	0.60505
	Test	2.5986	1.612	0.42733
	Validation	2.5599	1.6	0.3829

Figure 6 comparing the real target and the output of ANN model for “DBT” in four selected stocks.

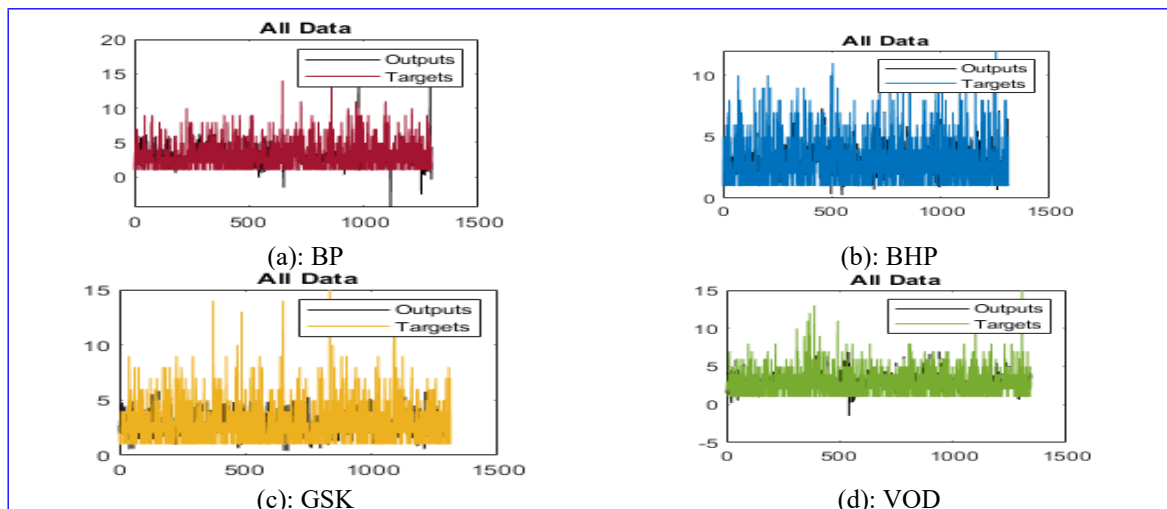


Figure 6: Comparing the actual and the predictions of the ANN model for DBT

4.7. DBPT Prediction Results

The results of predicting the distance between two consecutive peak and troughs (DBPT) are presented in Table 11.

Table 11: The results of DBPT Prediction using ANN model

Stock	Dataset	Evaluation Metrics		
		MSE	RMSE	R ²
BP	All	0.7905	0.8891	0.77848
	Train	0.73511	0.85739	0.80403
	Test	0.92596	0.96227	0.71201
	Validation	0.91368	0.95587	0.70632
BHP	All	0.74249	0.86168	0.7878
	Train	0.69034	0.83087	0.8003
	Test	1.044	1.0218	0.70591
	Validation	0.68441	0.82729	0.82143
GSK	All	0.8889	0.94282	0.7276
	Train	0.83886	0.91589	0.75708
	Test	0.9195	0.95891	0.62738
	Validation	1.0914	1.0447	0.67623
VOD	All	0.77027	0.87765	0.73033
	Train	0.59065	0.76854	0.79227
	Test	1.1288	1.0624	0.61827
	Validation	1.2502	1.1181	0.57893

Figure 7 presents the real target and the output of ANN for “DBPT” in four selected stocks.

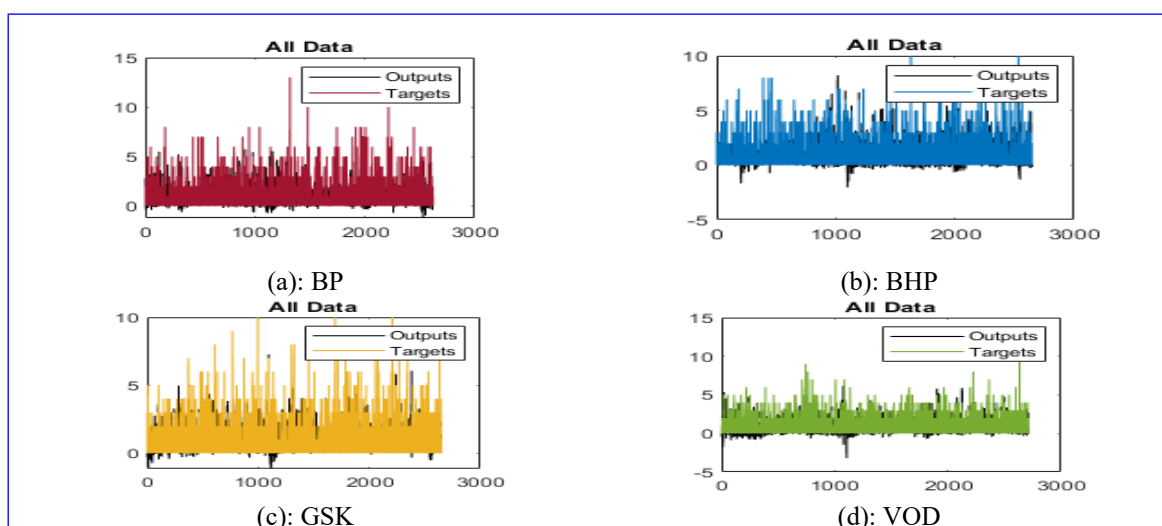


Figure 7: Comparing the actual and the predictions of the ANN model for DBPT

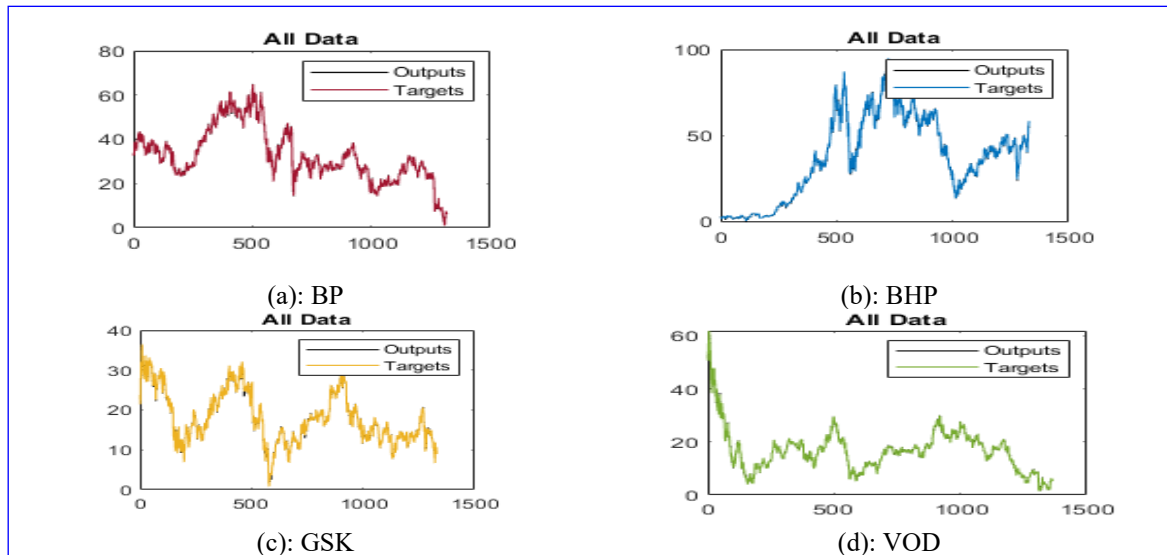
4.8. Peak Height Prediction

The result of predicting the height of the next peak (PKHe) is presented in Table 12. The R² values are greater than 0.9995 for all stocks in train, test, and validation data.

Table 12: The results of prediction of peak height using ANN model

Stock	Dataset	Evaluation Metrics		
		MSE	RMSE	R ²
BP	All	0.024125	0.15532	0.99992
	Train	0.018731	0.13686	0.99994
	Test	0.047378	0.21766	0.99982
	Validation	0.026013	0.16129	0.99992
BHP	All	0.098238	0.31343	0.99992
	Train	0.10059	0.31716	0.99992
	Test	0.083997	0.28982	0.99993
	Validation	0.10153	0.31864	0.99991
GSK	All	0.021662	0.14718	0.99976
	Train	0.010157	0.10078	0.99999
	Test	0.055278	0.23511	0.99942
	Validation	0.041544	0.20382	0.99952
VOD	All	0.016971	0.13027	0.99987
	Train	0.0038197	0.061804	0.99997
	Test	0.042791	0.20686	0.99962
	Validation	0.052353	0.22881	0.99969

Figure 8 presents the real target and the output of ANN model for “PKHe” in four selected stocks.

**Figure 8: Comparing the actual and the predictions of the ANN model for PKHe**

4.9. Trough Depth Prediction

The results of predicting the depth of the next trough (TRUDe) are presented in Table 13. The R² values are greater than 0.9996 for all stocks in train, test, and validation data.

Table 13: The results of prediction of trough depth using ANN model

Stock	Dataset	Evaluation Metrics		
		MSE	RMSE	R ²
BP	All	0.043989	0.20973	0.99984
	Train	0.043643	0.20891	0.99984
	Test	0.05226	0.22861	0.99981
	Validation	0.037325	0.1932	0.99988
BHP	All	0.081624	0.2857	0.99993
	Train	0.070385	0.2653	0.99994
	Test	0.11383	0.33739	0.9999

GSK	Validation	0.10173	0.31896	0.9999
	All	0.028403	0.16853	0.99965
	Train	0.027001	0.16432	0.99967
	Test	0.032709	0.18086	0.99962
VOD	Validation	0.030614	0.17497	0.99963
	All	0.030149	0.17363	0.99975
	Train	0.028467	0.16872	0.99977
	Test	0.032544	0.1804	0.9997
	Validation	0.035579	0.18862	0.99971

Figure 9 presents the real target and the output of ANN model for “TRUDE” in four selected stocks.

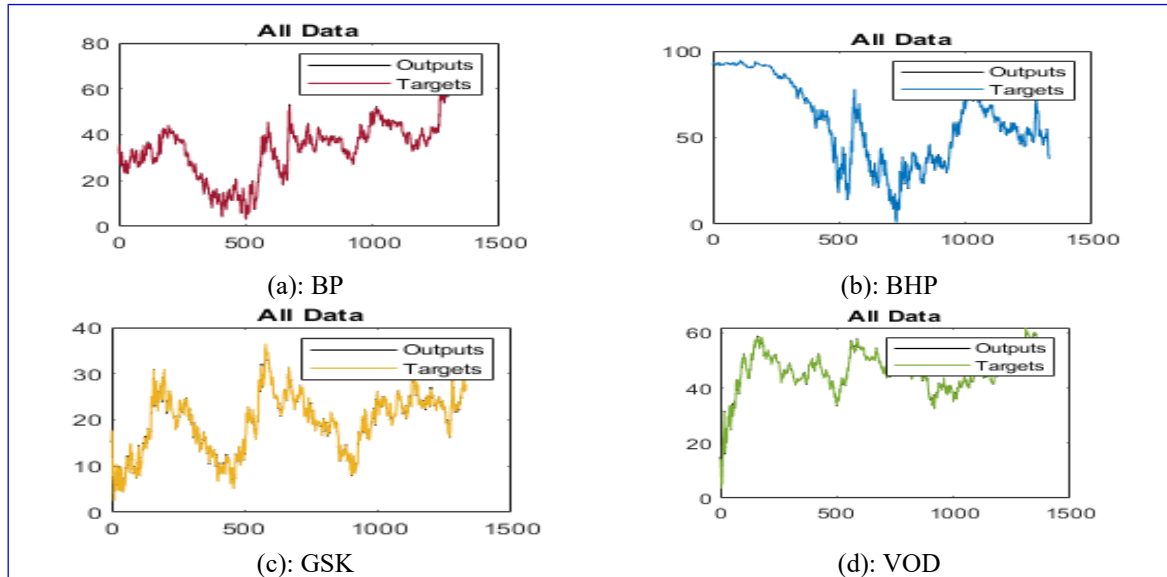


Figure 9: Comparing the actual and the predictions of the ANN model for TRUDE

4.10. Delta Prediction

Delta compares the closing stock price of day with the closing price of the previous trading day. The prediction performance of the proposed models in terms of MSE, RMSE, and R^2 , implemented on the 4 selected stocks are shown in Table 14. As can be seen, the prediction model has a good performance. The R^2 values are greater than 0.9998 for all stocks in train, test, and validation data.

Table 14: The results of Delta prediction using ANN model

Stock	Dataset	Evaluation Metrics		
		MSE	RMSE	R^2
BP	All Data	0.0001664	0.0129	0.99988
	Train	0.00012249	0.011067	0.99991
	Test	0.00036693	0.019155	0.99972
	Validation	0.00017089	0.013073	0.99989
BHP	All Data	7.3115e-05	0.0085507	0.99997
	Train	5.948e-05	0.0077123	0.99998
	Test	0.00011297	0.0106297	0.99996
	Validation	9.6921e-05	0.0098448	0.99996
GSK	All Data	1.1134e-05	0.0033367	0.99999
	Train	4.77e-06	0.002184	0.99999
	Test	1.8325e-05	0.0042807	0.99998
	Validation	3.3653e-05	0.0058011	0.99995

VOD	All Data	3.256e-05	0.0057061	0.99995
	Train	1.3412e-05	0.0036623	0.99989
	Test	0.00010458	0.010226	0.99998
	Validation	4.9939e-05	0.0070668	0.99993

Figure 10 comparing the real target and the output of proposed ANN for “Delta” in four selected stocks.

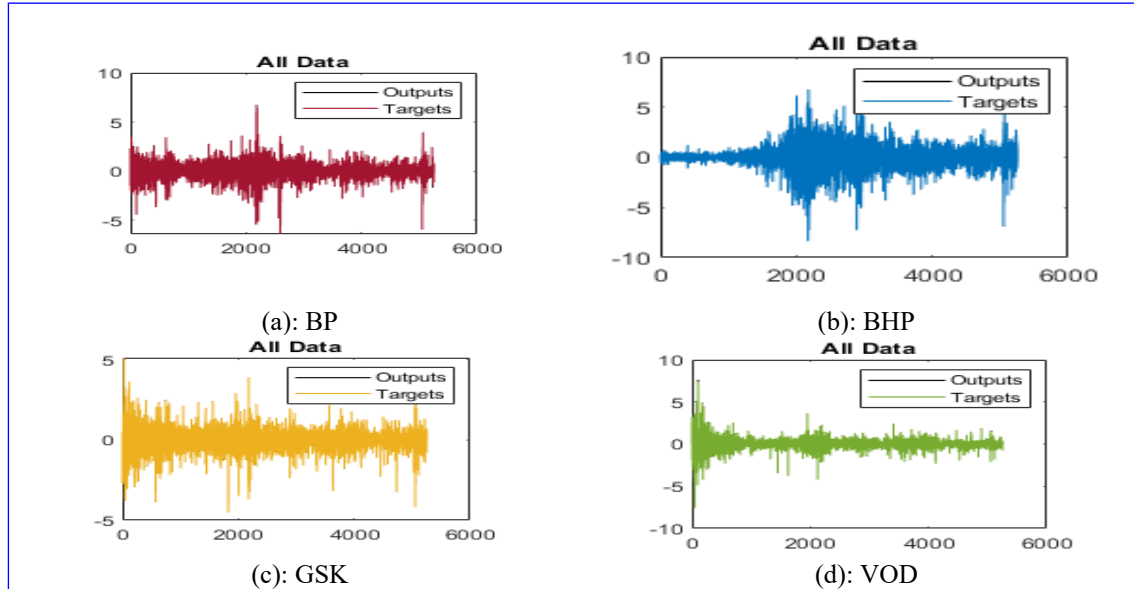


Figure 10: Comparing the actual and the predictions of the ANN model for Delta

As clear the output and target values are exactly the same for more than 99.998 percentage of data patterns.

4.11. Return Prediction

The results of stock return prediction are presented in [Table 15](#). As can be seen, the prediction model has a good performance for all selected stocks. The R^2 values are greater than 0.9992 for all stocks in train, test, and validation data.

Table 15: The results of return prediction using ANN model

Stock	Dataset	Evaluation Metrics		
		MSE	RMSE	R^2
BP	All Data	5.9796e-08	0.00024453	0.99992
	Train	6.2174e-09	7.885e-05	0.99999
	Test	3.1725e-07	0.00056325	0.99973
	Validation	5.249e-08	0.00022911	0.99993
BHP	All Data	1.3216e-08	0.0001496	0.99999
	Train	8.5859e-09	9.266e-05	0.99999
	Test	1.5154e-08	0.0001231	0.99999
	Validation	3.8754e-09	6.2253e-05	0.99999
GSK	All Data	7.5035e-08	0.00027393	0.99982
	Train	8.7082e-09	9.9318e-05	0.99998
	Test	1.9731e-07	0.00044419	0.99952
	Validation	2.6243e-07	0.00051228	0.99928
VOD	All Data	1.0908e-08	0.00010444	0.99999
	Train	9.8742e-09	9.9369e-05	0.99999
	Test	1.0343e-08	0.0001017	0.99999
	Validation	1.6297e-08	0.00012766	0.99998

Figure 11 comparing the real target and the output of proposed ANN for “Return” in four selected stocks.

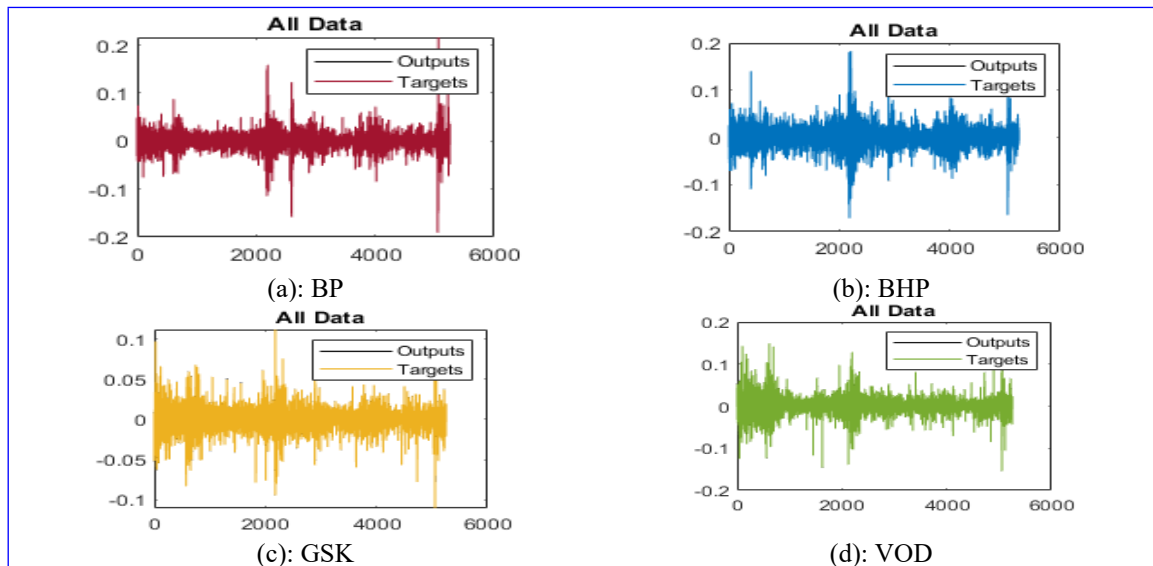


Figure 11: Comparing the actual and the predictions of the ANN model for Return

4.12 Validation of the proposed Prediction method

In order to check the validity of proposed prediction approach the BHP stock is selected.

The results of the prediction of continuous indices and classification of binary targets of BHP stock for the selected period are presented in subsection 4.2 to 4.11 .

Considering that the R2 values for validation data for DBP and DBT indices of BHP stock were about 34% and 42%, respectively, these indices were removed and validation was performed with the remaining eight indices. The prediction of Engineered and Derivative indices for the next 19 days (January 4, 2021, and January 29, 2021) for BHP stock has been accomplished using the proposed models. The results are presented in Table 16.

As can be seen, the original closing prices of the BHP stock for January 4th to January 29th are presented in the first column of Table 16.

Firstly, the performance of ‘Trend’, ‘PK’, and ‘TRU’ indices classification models have been evaluated.

- The original value of the trend index is also presented in the second column. The results of the classification ‘Trend’ is presented in the third column of Table 16. By comparing the original and predicted values of this index, it can be said that the model for the classification of this variable has performed well.
- The original and predicted values of the ‘PK’ target variable are also presented in the fourth and fifth columns of Table 16, respectively. As mentioned in section 4.3, the prediction model of this target variable has an accuracy of about 75.6%. It can be seen that this target variable has not been correctly predicted for some days.
- The original and predicted values of the ‘TRU’ target variable is presented in the sixth and seventh columns of Table 16, respectively. As can be seen, this variable was not correctly predicted on day and the prediction for the other days is done correctly and the model accurately predicts the troughs.

In the following, the validity of continuous indices forecasting models including ‘DBPT’, ‘PKHe’, ‘TRUDe’, ‘Delta’, and ‘Return’, is discussed.

- The original and predicted value of the DBPT index are presented in the Eighth and ninth columns of Table 16. According to the original value of PK index (column 4), The price is at the peak on day $t+5$. the distance from this peak to the next trough is equal to zero. The prediction value for the distance to the next peak is equal to 0.06 and therefore the prediction model had an acceptable performance on this day. Predictions for , , and were not accurate. As mentioned in Section 4.7, the variable prediction model of this target variable has a coefficient of determination of about 78.78%.

The original and predicted values of ‘PKHe’, ‘TRUDe’, ‘Delta’, and ‘Return’ indices are presented in columns 10 to 17 of Table 16. To determine whether there is any significant difference between the average of original values and predicted values of these indices, independent t-test have been used, and the test hypothesis considered as Eq. (20)

(20)

If the p-value in the output of the independent t-test is lower than the Type I error, the null hypothesis is rejected, meaning that there is a significant relationship between the means. Also, if one of the upper bound and the lower bound is positive and the other is negative, this means that the means of the two groups are not significantly different. The results of the t-test of each of these indices are presented in Table 17.

Table 16: The predicted values of the proposed indices for the new 19 days for BHP stock

Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
	Original Closing Price (CP1)	Original Trend	Predicted Trend	Original PK	Predicted PK	Original TRU	Predicted TRU	Original DBPT	Predicted DBPT	Original PKHe	Predicted PKHe	Original TRUDe	Predicted TRUDe	Original Delta	Predicted Delta	Original Return	Predicted Return	Calculated Closing Price using Predicted Delta (CP2)	Calculated Closing Price using Predicted Return (CP3)
Dec 31, 2020	65.34	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
t+1	67.35	1	1	0	1	0	0							2.01	2.017	0.031	0.031	67.357	67.366
t+2	69.5	1	1	0	1	0	0							2.150	2.155	0.0319	0.032	69.512	69.521
t+3	71.71	1	1	0	1	0	0							2.201	2.200	0.0318	0.032	71.712	71.746
t+4	73.18	1	1	0	1	0	0							1.470	1.471	0.0205	0.020	73.183	73.181
t+5	73.46	1	1	1	0	0	0	0		65.53	65.12			0.271	0.271	0.0038	0.004	73.454	73.474
t+6	71.28	0	0	0	0	1	1	0				31.4	31.94	-2.18	-2.157	-0.0296	-0.030	71.297	71.269
t+7	71.95	1	1	1	1	0	0	0		64.02	63.66			0.669	0.666	0.0094	0.009	71.963	71.911
t+8	71.07	0	0	0	0	1	1	0				31.61	31.88	-0.88	-0.885	-0.0122	-0.012	71.078	71.048
t+9	73.88	1	1	1	1	0	0	0		65.95	66.09			2.809	2.821	0.0395	0.040	73.899	73.890
t+10	70.61	0	0	0	0	1	1	1				32.07	31.16	-3.27	-3.289	-0.0443	-0.044	70.61	70.639
t+11	71.44	1	1	0	1	0	0							0.830	0.835	0.0117	0.012	71.445	71.486
t+12	72.78	1	1	1	1	0	0	2		64.85	65.45			1.339	1.346	0.0187	0.019	72.791	72.845
t+13	72.33	0	0	0	0	0	0							-0.45	-0.454	-0.00618	-0.006	72.337	72.407
t+14	71.79	0	0	0	0	0	0							-0.54	-0.541	-0.00747	-0.007	71.796	71.901
t+15	71.21	0	0	0	0	1	0	0				31.47	31.56	-0.58	-0.582	-0.00808	-0.008	71.214	71.325
t+16	71.8	1	1	1	1	0	0	0		63.87	64.18			0.590	0.592	0.00828	0.008	71.806	71.896
t+17	69.14	0	0	0	0	1	1	0				33.54	33.50	-2.66	-2.675	-0.03705	-0.037	69.131	69.236
t+18	69.31	1	1	1	0	0	0	0		61.38	61.99			0.169	0.166	0.00246	0.002	69.297	69.374
t+19	66.78	0	0	0	0	1	1	0				35.9	36.05	-2.53	-2.535	-0.0365	-0.036	66.762	66.877

As can be seen in Table 17, in all cases the p-value is greater than the Type I error, and also the lower bound is negative and the upper bound is positive, therefore the means of the two groups for each mentioned index are not significantly different.

Figure 12 presents the comparison of original and predicted values of these indices with a boxplot.

Table 17: Two Sample t-test results

Attribute	t	df	p-value	95 percent confidence interval		mean in group Original	mean in group Predicted
PKHe	-0.16515	9.89	0.8721	-2.152320	1.855653	64.26667	64.41500
TRUDe	-0.016011	9.89	0.9875	-2.336398	2.303065	32.66500	32.68167
Delta	-0.000363	36	0.9997	-1.176272	1.175851	0.07463158	0.07484211
Return	-0.008624	36	0.9932	-0.016657	0.016516	0.001456	0.001526

alternative hypothesis: true difference in means is not equal to 0

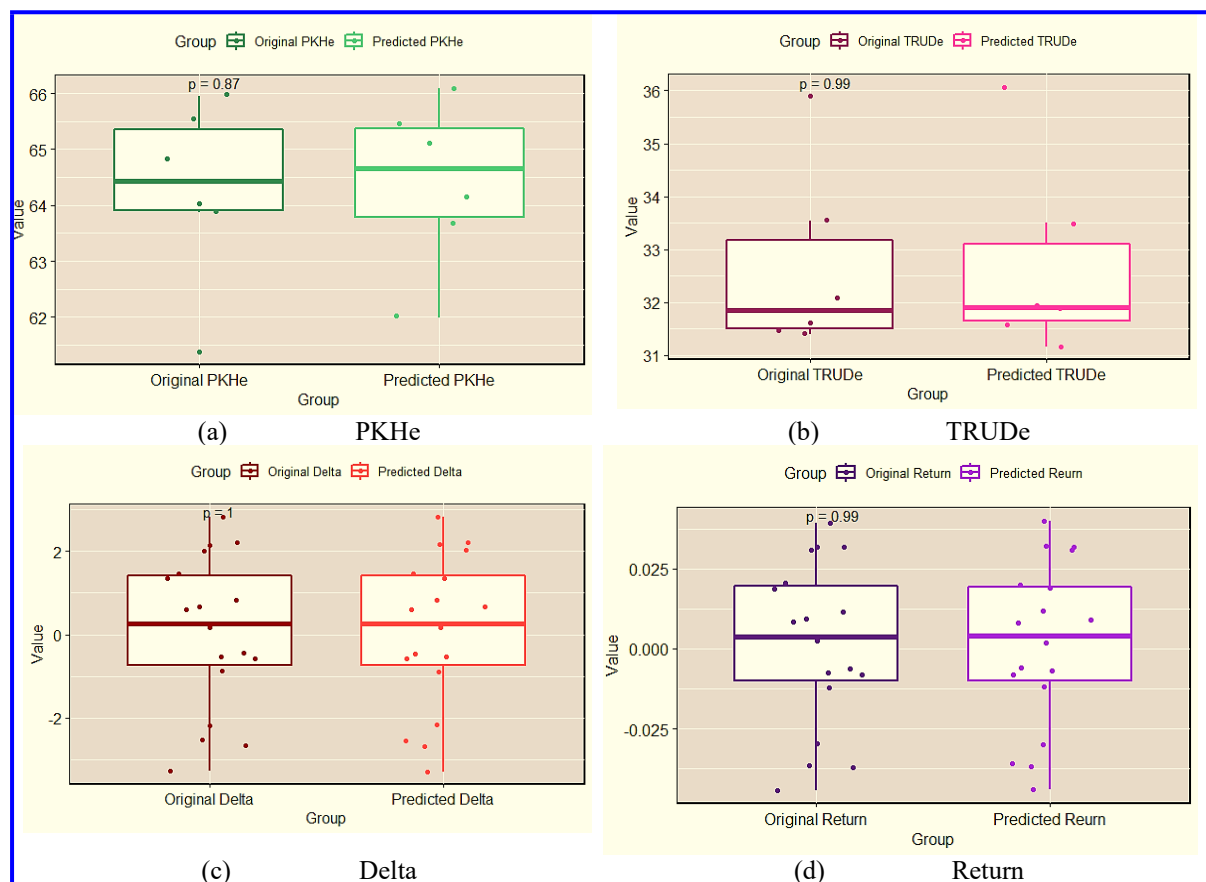


Figure 12: Comparison of original and predicted values with boxplot

In the following, the closing price for the selected period (January 4, 2021, and January 29, 2021) have been calculated using the predicted values of 'Delta' and 'Return'.

Considering the availability of the closing price on day t (65.34) and using the predicted value of the 'Delta' target variable (column 15) and Eq. (9), the closing price on day $t+1$ can be calculated as Eq. (21):

(21)

The closing price on day t can also be calculated according to the price calculated for day $t + 1$ using Eq. (9).
(22)

The closing price for the coming days is calculated in the same way. The calculated closing price using predicted 'Delta' (CP2) are presented in column 18 of Table 16.

The predicted value of the 'Return' is presented in the column 17 of Table 16. Once again, this time using Eq. (10), the closing price on day t can be calculated as Eq. (23):
(23)

The calculated Closing price using predicted 'Return' (CP3) are presented in column 19 of Table 16.

In order to compare the average of original closing price (CP1), calculated closing price using predicted 'Delta' (CP2), and calculated closing price using predicted 'Return' (CP3), independent t-test have been used, and the test hypothesis considered as Eq. (24).

At least between two groups
(24)

The results of the average of CP1, CP2, and CP3 are presented in Table 18. Figure 13 presents the comparison of the original and predicted closing prices using a boxplot. As can be seen, there is no significant difference between the means according to P-values. CP1, CP2, and CP3 are provided in Figure 14 for the next 19 days. It is clear that there are similarities.

Table 18: Two Sample t-test results for comparing original and predicted closing prices

Attribute	t	df	p-value	95 percent confidence interval		mean in group Original	mean in group Predicted
CP1-CP2	36	0.0062259	0.9951	-1.264825	1.272615	71.08653	71.08263
CP1-CP3	36	0.06944	0.945	-1.220309	1.306835	71.12589	71.08263
CP2-CP3	36	-0.063106	0.95	-1.304587	1.225850	71.08653	71.12589

alternative hypothesis: true difference in means is not equal to 0

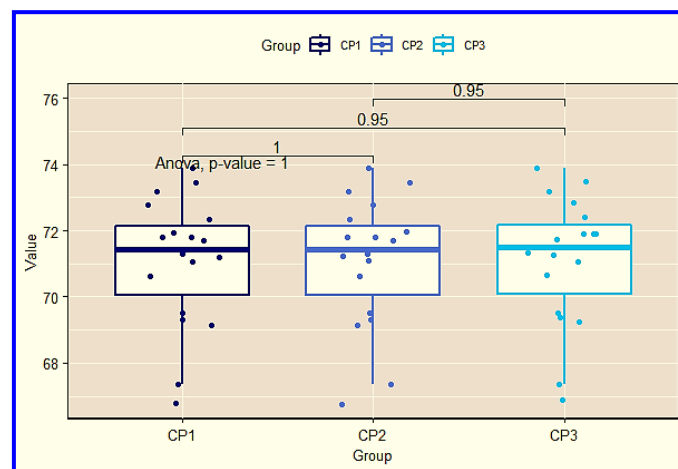


Figure 13: Comparison of original and predicted values of closing price with boxplot

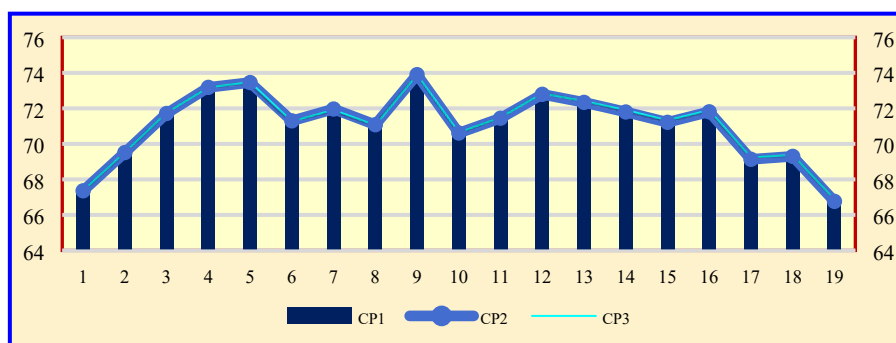


Figure 14: original and predicted values of closing price for the next 19 days

5. Conclusion Remarks and Suggestion for Future Researches

Accurate estimation of future stock prices can be of significant benefit to investors in the field of stock exchanges. In this paper, a multi-stage machine learning approach was proposed to predict the next day's stock prices. The proposed approach had three main stages, i.e., data collection, feature space generation, and prediction. In the first stage, 5283 records of daily historical series of open prices, close prices, the highest prices, the lowest prices, and the trading volume of assets of four well-known stocks were selected from FTSE 100. In the second stage, a novel set of engineered and derivative indices extracted from the original time series for stock price prediction were proposed. In the third stage, two Multi-Layer Perceptron (MLP) were proposed to predict the next day's stock prices using the engineered discrete and continuous indices.

The proposed approach in this paper helps the investor in predicting the stock price by presenting a set of new indices extracted from the closing price of stocks. There are several one-day-ahead targets that can be interesting for investors. For example, forecasting the amount of increase of the closing price in the next day, the return of the next day, the price trend in the next day. In addition to these well-known indicators, some novel indices such as determining whether the price will be at its peak the next day or not? predicting whether the price will be in the trough the next day or not? predicting the height of the next peak, predicting the depth of the next trough, predicting the distance from one peak to the next peak, distance from one trough to the next trough, the distance of one peak from the next trough and vice versa, were proposed and developed in this research. These novel indices can be very attractive for the investor. We called such indicators engineered and derivative indices. The aforementioned indices were extracted from daily stock price data to be considered as target variables. In fact, in this research, instead of directly predicting stock prices, engineered indices were used.

Ten different day-ahead indices were defined in order to predict these indices. A set of input variables including several lagged variables of returns, and several technical indicators were considered. The proposed approach is implemented on 4 stocks namely "BP", "BHP", "GSK", and "VOD" that were selected from the daily stock exchange of FTSE 100 between January 1, 2000, and December 31, 2020. Results revealed that taking into consideration various predictive factors and a novel set of target variables could be effective in predicting the stock price.

The proposed approach can be improved in the following aspects in future research. Some fundamental variables can also be considered as the input variables. Also, some base indicators such as gold, oil, gas prices, interest and inflation rates, import and export rates, and

GDP may be considered in future research. The proposed approach in this research can be used to the preselection of stocks to form a portfolio. Therefore, combining this approach with portfolio optimization problems may be interesting in future research.

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